OPERATING SYSTEM INFORMATION

Windows, Linux, macOS Installation and OS Settings

Introduction

An operating system (OS) is software that manages a computer's hardware and applications. Examples are Windows, Linux, and macOS. This assignment explains how to check system information, prepare Linux installation media, change macOS settings, create a Linux user, and understand Windows 10 installation.

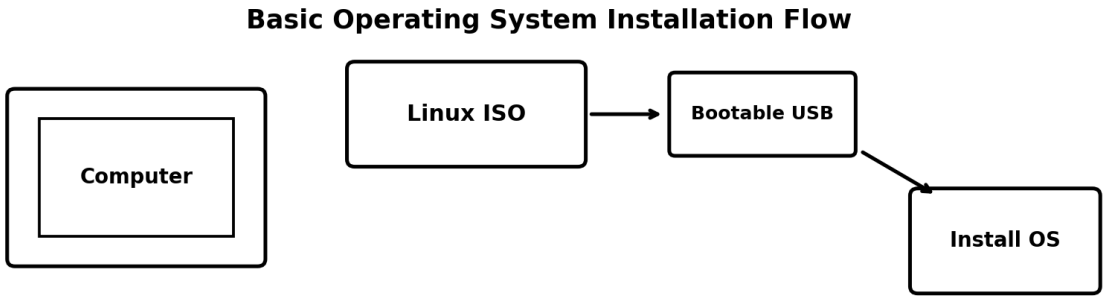


Figure: Basic steps for preparing and installing an operating system.

Task 1: Check Windows System Information

Aim: To view the Windows version, processor, and RAM details.

Steps:

1. Press Windows + R on the keyboard.
2. Type msinfo32 and press Enter.
3. The System Information window will open.
4. On System Summary, find OS Name and Version.
5. Find Processor to see processor information.
6. Find Installed Physical Memory (RAM) to see the RAM size.
7. Take a screenshot with Windows + Shift + S and save it for the assignment.

Result: The Windows version, processor, and installed RAM can be seen in System Information.

Screenshot: Insert your own screenshot here after completing the steps.

Task 2: Create a Bootable Linux USB

Aim: To prepare a USB drive that can start a Linux installation.

Steps:

8. Visit the official Ubuntu website at ubuntu.com/download/desktop or Fedora at getfedora.org and download the ISO file.
9. Use a USB drive with at least 8 GB of space. Back up its files first because creating a bootable USB erases the drive.
10. On Windows, download Rufus from its official website, rufus.ie.
11. Connect the USB drive and open Rufus.

12. Choose the USB drive under Device and choose the Linux ISO under Boot selection.
13. Keep the recommended settings unless your computer needs different settings.
14. Click Start and confirm the warning that the USB contents will be erased.
15. Wait until Rufus reports that the process is complete, then safely eject the USB.

Result: The USB drive is ready to boot and start the Linux installer.

Safety note: Select the correct USB drive. Choosing the wrong drive can erase important files. If you do not have a spare USB, use VirtualBox and attach the ISO to a virtual machine.

Task 3: Change Wallpaper and Language on macOS

Aim: To change the desktop wallpaper and add Hindi or Gujarati as a language.

Steps:

16. Open the Apple menu and choose System Preferences. On newer macOS versions, this may be called System Settings.
17. Open Desktop & Screen Saver or Wallpaper, depending on the macOS version.
18. Choose a wallpaper from the available pictures.
19. Return to System Preferences or System Settings and open Language & Region.
20. Click the plus (+) button to add a language, then choose Gujarati or Hindi.
21. If asked whether to use the new language as the primary language, choose your preferred option. Some versions may require apps to be restarted.
22. Check that the selected language appears in the language list.

Result: The wallpaper is changed and Hindi or Gujarati is added to Language & Region.

Note: Menu names vary by macOS version. If a Mac is not available, use an authorised macOS virtual machine if permitted by the instructor.

Task 4: Install Ubuntu and Create the 'gamer' User

Aim: To install Ubuntu, create a user named gamer, and enable automatic login.

Steps:

23. Download the Ubuntu Desktop ISO from the official Ubuntu website.
24. Back up important files before changing partitions or installing an operating system.
25. Start the computer from the bootable USB, or attach the ISO to a virtual machine such as VirtualBox.
26. Choose Try or Install Ubuntu, then select Install Ubuntu.
27. Follow the instructions for language, keyboard, network, and installation type.
28. Choose the correct target disk carefully. In a virtual machine, use its virtual disk. Selecting the wrong disk can erase data.
29. When asked for account details, enter the username gamer and create a strong password.
30. Complete the installation and restart when prompted. Remove the USB if using a physical computer.
31. Open Settings and find System or Users, depending on the Ubuntu version.
32. Unlock the settings if required, select the gamer account, and turn on Automatic Login if the option is available.
33. Sign out or restart to verify the login behaviour. Take a screenshot showing the username or account settings.

Result: Ubuntu is installed and the gamer account is created. Automatic login is enabled if supported by the selected Ubuntu version.

Security note: Automatic login lets anyone with physical access open the account without entering its password. Use it only on a safe personal or virtual machine.

Task 5: Step-by-Step Guide to Install Windows 10

Aim: To understand Windows 10 installation and adjust privacy and update settings.

Important: Standard Windows 10 support ended on 14 October 2025. For a computer used online, consider a supported operating system or check whether the device is eligible for Microsoft's Extended Security Updates (ESU). If Windows 10 is required for this assignment, use a licensed copy and understand the security risk.

A. Before Installation

34. Back up documents, photos, and other important files to an external drive or trusted cloud storage.
35. Check the laptop maker’s website for Windows drivers and confirm that the device supports Windows 10.
36. Keep the product key or digital licence information available if required.
37. Download Windows installation media from Microsoft's official Windows 10 download page.
38. Create a bootable USB with Microsoft's Media Creation Tool or another trusted tool. The USB drive will be erased.

B. Install Windows 10

39. Connect the bootable USB to the laptop and restart it.
40. Open the boot menu using the key shown by the laptop maker, then select the USB drive.
41. Choose language, time, and keyboard settings, then click Next and Install now.
42. Enter the product key if asked. Choose the Windows edition that matches your licence.
43. Choose Custom: Install Windows only (advanced) for a clean installation.
44. At the partition screen, identify the correct target drive carefully. Deleting or formatting partitions removes data.
45. Only delete partitions on the intended target drive if you have a verified backup and are certain it is the correct drive. Otherwise stop and ask the instructor or a technician.
46. Select the intended unallocated space or target partition and continue.
47. Follow the setup screens to choose region, keyboard, network, and account settings.
48. After setup, install required drivers from the laptop maker’s official website and restart if prompted.

C. Privacy and Updates After Installation

49. Open Settings > Update & Security > Windows Update and check for available updates. Install applicable updates and restart if requested.
50. Open Settings > Privacy and review permissions for location, camera, microphone, contacts, and other personal data. Turn off access you do not need.
51. Read privacy choices carefully during setup and turn off optional data sharing or personalised experiences if you do not want them.
52. Use a strong password or PIN and keep Windows Security protection enabled.
53. Check Device Manager for warning icons and install drivers from the computer manufacturer’s official website.

D. Steps That May Need Clarification

The boot-menu key, partition layout, product-key prompt, and driver installation can differ by computer. The exact screens depend on the laptop model and installer version. If a partition or disk is unclear, do not guess or delete it; ask the instructor or a technician.

Result: The Windows 10 installation process and important partitioning, privacy, and update settings are understood.

Conclusion

In this assignment, I learned how to check Windows system information, prepare a bootable Linux USB, change macOS wallpaper and language settings, create an Ubuntu user named gamer, and understand Windows 10 installation. Backing up files, choosing the correct disk, using official downloads, and checking security settings are important during OS installation.